Literature Review
Detailed Paper Analysis

DSCI-41 | Traffic Volume Estimation & Congestion Forecasting

16 Papers | Sorted by Relevance | With Research Gaps & Improvements

Each paper includes:

- Published Date & Venue
- Abstract & Introduction
- Technologies Used
- Conclusion
- Research Gap Identified
- Our Proposed Improvements

1

Traffic Impact Assessment System using YOLOv5 and ByteTrack

Authors: Jing Jie Ng, Kah Ong Michael Goh, Connie Tee

Published: September 2023 | Journal of Informatics and Web Engineering, Vol. 2, No. 2, pp. 99-113

https://www.researchgate.net/publication/373895809

Abstract

This paper addresses the need for efficient, low-cost traffic monitoring software. The authors propose a system that utilizes YOLOv5 for vehicle detection and ByteTrack for multi-object tracking. The system supports vehicle detection and classification, measuring vehicle idling time by region, counting vehicles at specific junctions, and allowing users to draw regions and counting lines on video feeds for precise monitoring.

Introduction

Traffic congestion is a critical problem in Malaysia and other developing nations. Traditional traffic monitoring relies on expensive hardware sensors that are difficult to maintain. The authors propose a computer vision-based approach using modern deep learning object detection (YOLOv5) combined with a state-of-the-art tracking algorithm (ByteTrack) to create a cost-effective, real-time traffic assessment system.

Technologies Used

YOLOv5 (object detection), ByteTrack (multi-object tracking), StrongSORT and OC-SORT (comparison trackers), Python, OpenCV, custom GUI for region/line drawing on camera feeds.

Conclusion

The proposed system effectively serves as a viable, low-cost alternative to traditional hardware-based traffic monitoring methods. ByteTrack demonstrated superior speed and accuracy compared to StrongSORT and OC-SORT. The synergy between YOLOv5 and ByteTrack offers a robust solution for real-time traffic assessment with potential to enhance intelligent traffic control systems.

Research Gap

1) Uses YOLOv5 which is now outdated; YOLOv8's anchor-free architecture offers better accuracy on small and occluded vehicles. 2) No weather robustness evaluation - system tested only under clear conditions. 3) No forecasting component - system counts vehicles but does not predict future congestion. 4) Uses hard counting (binary integer counts) which provides no uncertainty information about detection quality. 5) No evaluation on standardized MOT benchmarks (MOTA, IDF1).

Our Proposed Improvements

1) Upgrade from YOLOv5 to YOLOv8 for better anchor-free detection of mixed vehicle sizes. 2) Add adaptive weather preprocessing (CLAHE for night, dehazing for fog) to sustain accuracy under adverse conditions. 3) Extend the pipeline with a forecasting module (LSTM/XGBoost) that takes vehicle counts and predicts congestion 15-30 minutes ahead. 4) Replace hard counting with confidence-weighted soft counting to represent detection uncertainty. 5) Evaluate on standard benchmarks with MOTA and IDF1 metrics.

2

Barbados Traffic Analysis Challenge

Authors: Zindi Africa

Published: 2024 | Zindi Competition Documentation

https://zindi.africa/competitions/barbados-traffic-analysis-challenge

Abstract

The Barbados Traffic Analysis Challenge provides a dataset of video streams captured by four cameras positioned at different entrances and exits of the Norman Niles roundabout. Participants must predict traffic congestion levels (entrance and exit ratings) 15 minutes into the future based on historical video data.

Introduction

The challenge aims to help the Ministry of Transport and Works identify the root causes of traffic flow issues in Barbados. The core task involves extracting features from unstructured video data (approximately 575 GB) to determine congestion, simulating a real-world scenario where past video must predict future traffic conditions.

Technologies Used

Video stream processing, feature extraction, time-series forecasting, macro-F1 and accuracy evaluation metrics, Google Cloud Storage.

Conclusion

The challenge highlights the difficulty of predicting congestion from raw video feeds without explicit object detection/tracking labels. The winning solutions must bridge the gap between unstructured video pixels and structured traffic flow prediction.

Research Gap

1) Competition dataset provides only congestion ratings, no bounding box annotations for training object detectors. 2) 15-minute prediction window is fixed, not flexible for shorter (5 min) or longer (30 min) horizons. 3) Does not explicitly provide weather condition labels alongside the video. 4) Evaluation metrics (Macro-F1/Accuracy) treat congestion as discrete classes rather than a continuous volume regression. 5) Does not account for camera movement or perspective changes.

Our Proposed Improvements

1) Use pre-trained YOLOv8 to automatically generate vehicle counts, effectively creating the missing structured data. 2) Build a flexible forecasting model that can predict at multiple time horizons. 3) Train a lightweight weather classifier to tag frames and apply appropriate preprocessing before detection. 4) Output continuous traffic volume estimates alongside discrete congestion classes for better traffic analysis. 5) Implement stabilization and perspective correction in the preprocessing pipeline.

3 ByteTrack: Multi-Object Tracking by Associating Every Detection Box

Authors: Yifu Zhang, Peize Sun, Yi Jiang, Dongdong Yu, Fucheng Weng, Zehuan Yuan, Ping Luo, Wenyu Liu, Xinggang Wang
Published: October 2022 | European Conference on Computer Vision (ECCV) 2022
https://arxiv.org/abs/2110.06864

Abstract

Most MOT methods obtain identities by associating only high-score detection boxes, discarding low-score detections that often contain occluded objects. ByteTrack proposes associating every detection box instead of only high-score ones. By leveraging the similarities between low-score detection boxes and tracklets, the method recovers true objects while filtering out background noise. Applied to nine different trackers, BYTE consistently improves IDF1 scores by 1-10 points.

Introduction

Multi-object tracking aims to estimate bounding boxes and identities of objects in video. The dominant paradigm is tracking-by-detection. Most methods discard low-confidence detections, losing information about partially occluded objects. ByteTrack challenges this by associating every detection box through a two-step matching process.

Technologies Used

YOLOX (detector), Kalman Filter (motion model), Hungarian Algorithm (data association), IoU-based matching, two-step association strategy (high-confidence first, then low-confidence), PyTorch.

Conclusion

Simple data association strategies that account for low-confidence detections significantly enhance tracking consistency. ByteTrack achieved 80.3 MOTA, 77.3 IDF1 on MOT17, and state-of-the-art on MOT20, HiEve, and BDD100K at 30 FPS on V100 GPU.

Research Gap

1) Relies purely on IoU-based motion association with no appearance features, which can cause identity switches when vehicles cross paths at intersections. 2) Evaluated primarily on pedestrian tracking benchmarks (MOT17/MOT20), not specifically on traffic/vehicle scenarios. 3) No handling of camera perspective distortion - vehicles far from camera have smaller IoU values. 4) No weather-specific evaluation - tracking performance under rain, fog, or night conditions is unknown. 5) Does not output confidence-weighted counts for downstream analytics.

Our Proposed Improvements

1) Integrate ByteTrack with YOLOv8 specifically for vehicle classes (cars, buses, trucks, motorcycles) and evaluate on traffic-specific datasets (Barbados Traffic). 2) Add homography-based perspective correction before IoU computation to handle camera angle distortion. 3) Test tracking robustness under adverse weather using ACDC/BDD100K conditions. 4) Feed ByteTrack's per-track confidence scores into a soft counting mechanism instead of binary counts. 5) Combine with a lightweight Re-ID feature for intersections where vehicles frequently cross paths.

4 Ultralytics YOLOv8

Authors: Glenn Jocher, Ayush Chaurasia, Jing Qiu

Published: January 2023 | GitHub Repository: ultralytics/ultralytics

https://github.com/ultralytics/ultralytics

Abstract

YOLOv8 is the latest iteration of the YOLO family of real-time object detectors. It introduces an anchor-free detection head, a decoupled head architecture separating classification and regression tasks, and the C2f module for improved feature extraction. YOLOv8 supports detection, segmentation, classification, and pose estimation within a unified framework.

Introduction

Building on the success of YOLOv5, YOLOv8 addresses the limitations of anchor-based detection by adopting an anchor-free approach. The decoupled head independently processes objectness, classification, and regression, leading to better convergence. The architecture provides multiple model sizes (n, s, m, l, x) to balance speed and accuracy.

Technologies Used

Anchor-free detection, C2f module, Decoupled Head, CSPDarknet backbone, Feature Pyramid Network (FPN) with Path Aggregation Network (PAN), CIoU and DFL loss functions, Mosaic augmentation, PyTorch, ONNX/TensorRT export.

Conclusion

YOLOv8 achieves 53.9% mAP on COCO at approximately 80 FPS, offering the best speed-accuracy trade-off in the YOLO family. Its anchor-free design improves detection of objects at unusual scales, making it suitable for traffic scenarios with mixed vehicle sizes.

Research Gap

1) Trained on COCO dataset which has limited traffic-specific classes - does not distinguish between vehicle subtypes relevant to traffic analysis. 2) No built-in mechanism for handling adverse weather degradation. 3) Detection confidence scores are not utilized for downstream soft counting. 4) Performance degrades significantly on small or heavily occluded vehicles at far distances. 5) No integration with tracking or counting - detection only.

Our Proposed Improvements

1) Fine-tune YOLOv8 on the Barbados Traffic Dataset with traffic-specific classes (cars, buses, trucks, motorcycles). 2) Apply domain-specific augmentation (synthetic rain, fog, darkness) during training to improve weather robustness. 3) Use detection confidence scores to generate soft counts instead of hard binary counts. 4) Integrate directly with ByteTrack for end-to-end detection-tracking-counting pipeline. 5) Apply CLAHE/dehazing preprocessing before inference on degraded frames.

5 Simple Online and Realtime Tracking with a Deep Association Metric (DeepSORT)

Authors: Nicolai Wojke, Alex Bewley, Dietrich Paulus

Published: 2017 | IEEE International Conference on Image Processing (ICIP) 2017

https://arxiv.org/abs/1703.07402

Abstract

This paper integrates appearance information to improve SORT's performance. The extension enables tracking through longer periods of occlusion, reducing identity switches by 45%. Computational complexity is shifted to an offline pre-training stage where a deep association metric is learned. During online application, associations use nearest neighbor queries in visual appearance space.

Introduction

The original SORT algorithm uses Kalman filtering and Hungarian algorithm but relies solely on motion cues, causing frequent identity switches during occlusions. DeepSORT adds a deep appearance descriptor (CNN trained on re-identification data) to enable re-identification after temporary occlusions. The cost combines Mahalanobis distance (motion) and cosine distance (appearance).

Technologies Used

Kalman Filter, Hungarian Algorithm, Deep CNN (trained on Mars re-identification dataset), Mahalanobis distance, Cosine distance, Cascade matching strategy, IoU association.

Conclusion

DeepSORT significantly reduces identity switches (45% reduction) while maintaining real-time performance. The design shifts computational burden to offline pre-training. However, the appearance model adds overhead compared to pure motion-based trackers.

Research Gap

1) The appearance model (Re-ID CNN) adds significant computational overhead, reducing FPS in dense traffic scenarios. 2) Re-ID model trained on person re-identification data, not vehicle re-identification - visual features may not transfer well to vehicles. 3) Does not handle low-confidence detections (unlike ByteTrack), discarding potentially useful occluded detections. 4) Fixed association thresholds do not adapt to varying camera conditions. 5) No integration with counting or forecasting systems.

Our Proposed Improvements

1) Use ByteTrack's two-step association (handling all detections) instead of DeepSORT's single-step matching for better occlusion handling in traffic. 2) If Re-ID is needed, train on vehicle-specific re-identification datasets like VeRi-776. 3) Combine DeepSORT's appearance features with ByteTrack's motion-first matching for a hybrid approach at intersections. 4) Feed tracking output into a counting module with virtual lines and then into a forecasting pipeline.

6 You Only Look Once: Unified, Real-Time Object Detection (YOLO)

Authors: Joseph Redmon, Santosh Divvala, Ross Girshick, Ali Farhadi

Published: June 2016 | IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2016

https://arxiv.org/abs/1506.02640

Abstract

YOLO frames object detection as a single regression problem, predicting bounding boxes and class probabilities from full images in one evaluation. The base network runs at 45 FPS and Fast YOLO at 155 FPS. While YOLO makes more localization errors, it is less likely to predict false positives on background and generalizes well to new domains.

Introduction

Traditional detection systems use complex, multi-stage pipelines (e.g., R-CNN) requiring thousands of evaluations per image. YOLO's core philosophy is to perform detection in a single pass, treating the entire pipeline as a single neural network optimized end-to-end.

Technologies Used

Darknet framework (C/CUDA), 24 convolutional layers + 2 fully connected layers, 7x7 grid prediction, Non-Maximum Suppression (NMS), unified regression-based detection, leaky ReLU.

Conclusion

YOLO demonstrates that detection can be performed in a single, fast forward pass. It is faster, simpler to train, and learns general object representations, making it ideal for real-time applications.

Research Gap

1) The original 7x7 grid prediction severely limits detection of small objects and nearby objects. 2) High localization error compared to two-stage detectors like Faster R-CNN. 3) Cannot handle varying object scales well within a single image. 4) Anchor-based design requires manual anchor definition for different datasets. 5) Trained on general datasets, not traffic-specific.

Our Proposed Improvements

1) Use YOLOv8 (the latest evolution) which resolves all these limitations with anchor-free detection, multi-scale FPN, and the C2f module. 2) Fine-tune on traffic-specific datasets for domain adaptation. 3) Combine with ByteTrack for tracking (original YOLO had no tracking capability). 4) The evolution from YOLOv1 to YOLOv8 demonstrates the architectural improvements our project leverages.

7

Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks

Authors: Shaoqing Ren, Kaiming He, Ross Girshick, Jian Sun

Published: 2015 | Advances in Neural Information Processing Systems (NeurIPS) 2015

https://arxiv.org/abs/1506.01497

Abstract

This paper introduces a Region Proposal Network (RPN) that shares convolutional features with the detection network, enabling nearly cost-free region proposals. For VGG-16, the system achieves 5 FPS on GPU with state-of-the-art accuracy on PASCAL VOC and MS COCO.

Introduction

Prior methods exposed region proposal computation as a bottleneck. The authors propose merging region proposal generation and detection into a single unified network by sharing convolutional features, creating an attention mechanism that tells the network where to look.

Technologies Used

Region Proposal Network (RPN), VGG-16 and ZF-Net backbones, anchor boxes, shared convolutional features, alternating training, Non-Maximum Suppression, Caffe framework.

Conclusion

The RPN eliminates the region proposal bottleneck. The unified architecture achieves state-of-the-art accuracy at 5 FPS. Faster R-CNN became the foundation for winning entries in ILSVRC and COCO 2015.

Research Gap

1) Processing speed of only 5 FPS is completely insufficient for real-time traffic video analysis (requires 25-30 FPS). 2) Two-stage architecture is inherently slower than single-stage detectors. 3) Complex training pipeline with alternating optimization. 4) High memory requirements due to region proposals. 5) Not designed for video - processes each frame independently with no temporal connection.

Our Proposed Improvements

1) Replace Faster R-CNN with YOLOv8 which achieves comparable accuracy at 80 FPS (16x faster). 2) Use Faster R-CNN only as an offline accuracy baseline to validate YOLOv8's detection quality. 3) Add temporal connection through ByteTrack tracking which Faster R-CNN completely lacks. 4) Our project demonstrates that single-stage detectors have closed the accuracy gap while maintaining real-time speed.

8

ACDC: The Adverse Conditions Dataset with Correspondences

Authors: Christos Sakaridis, Dengxin Dai, Luc Van Gool

Published: October 2021 | IEEE/CVF International Conference on Computer Vision (ICCV) 2021

https://arxiv.org/abs/2104.13395

Abstract

ACDC provides 4,006 images equally distributed across four adverse conditions: fog, nighttime, rain, and snow. Every adverse-condition image is paired with a corresponding normal-condition image. High-quality pixel-level semantic annotations and uncertainty masks are provided.

Introduction

Existing driving datasets focus on clear daytime conditions. Models trained on such data suffer significant performance degradation in adverse weather. ACDC addresses this with paired images enabling systematic study of condition-specific perception degradation.

Technologies Used

Pixel-level semantic segmentation annotations (19 classes), paired image correspondences, binary uncertainty masks, Cityscapes label format, evaluated with DeepLabv3+, DAFormer.

Conclusion

ACDC demonstrates that adverse conditions pose significant challenges to state-of-the-art methods. The paired correspondences enable unique research in domain adaptation and condition-robust perception.

Research Gap

1) Dataset is designed for semantic segmentation, not object detection - no bounding box annotations for vehicles. 2) Only 4,006 images which is relatively small for deep learning training. 3) Does not include traffic-specific scenarios like congested intersections or roundabouts. 4) No video sequences - only static images, so tracking cannot be evaluated. 5) No temporal or traffic flow data for forecasting evaluation.

Our Proposed Improvements

1) Use ACDC images for domain adaptation and data augmentation rather than primary training. 2) Apply synthetic weather transforms from ACDC conditions onto our Barbados Traffic video frames to create weather-degraded training data. 3) Use the paired normal/adverse images to quantify the exact mAP drop of our YOLOv8 model under each weather condition. 4) Build an adaptive preprocessing pipeline that automatically selects CLAHE (night), dehazing (fog), or contrast enhancement (rain) based on detected weather condition.

9 BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning

Authors: Fisher Yu, Haofeng Chen, Xin Wang, et al.

Published: 2020 | IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2020

https://arxiv.org/abs/1805.04687

Abstract

BDD100K is the largest open driving video dataset with 100,000 videos covering diverse geographic, environmental, and weather conditions. It supports multiple tasks: image tagging, lane detection, drivable area segmentation, object detection, semantic/instance/panoptic segmentation, and multi-object tracking.

Introduction

Autonomous driving requires robust perception across diverse real-world conditions. BDD100K provides 100K video clips covering diverse locations, times of day, and weather conditions, providing data to train models that generalize to the real world.

Technologies Used

Multi-task annotation pipeline, 100K video clips (40 seconds each, 720p, 30 FPS), 2D bounding box annotations, lane marking annotations, drivable area segmentation, semantic segmentation with 40 categories, MOT annotations.

Conclusion

BDD100K's scale and diversity make it an essential benchmark for developing robust driving perception systems. The heterogeneous multi-task annotations enable research into shared representations across tasks.

Research Gap

1) Annotations are from dashboard cameras (ego-centric view), not roadside/surveillance cameras which have a fundamentally different perspective. 2) Weather labels are coarse (day/night/rain) without fine-grained severity levels. 3) No traffic congestion labels or congestion severity ratings. 4) Vehicle tracking annotations are sparse and limited to key frames, not continuous. 5) No integration between detection/tracking and traffic forecasting tasks.

Our Proposed Improvements

1) Use BDD100K's weather-diverse images to augment our training data for weather robustness, applying perspective transforms to simulate roadside camera views. 2) Extract weather condition labels from BDD100K to train a lightweight weather classifier that selects condition-specific preprocessing. 3) Add congestion-level annotations to our Barbados dataset based on vehicle density thresholds. 4) Connect the detection-tracking pipeline to a forecasting module, which BDD100K does not address.

10

Single Image Haze Removal Using Dark Channel Prior

Authors: Kaiming He, Jian Sun, Xiaoou Tang

Published: 2009 | IEEE CVPR 2009

https://ieeexplore.ieee.org/document/5567108

Abstract

This paper proposes the dark channel prior for single image haze removal. Based on the observation that most local patches in haze-free images contain very low intensity pixels in at least one color channel, the method estimates haze thickness and recovers high-quality haze-free images.

Introduction

Haze reduces contrast and shifts colors, degrading computer vision algorithms. The dark channel prior is a statistical property of haze-free images that enables estimating atmospheric light and transmission maps from a single image.

Technologies Used

Dark Channel Prior, atmospheric scattering model, soft matting for transmission map refinement, guided image filtering, atmospheric light estimation, single-image dehazing.

Conclusion

The dark channel prior is a powerful method for single-image dehazing. It produces high-quality results without requiring multiple images or depth information and has become a foundational reference for visibility enhancement.

Research Gap

1) The method is computationally expensive due to soft matting, making it slow for real-time video processing (traffic CCTV at 25-30 FPS). 2) Assumes outdoor scenes with natural haze - may not work well with artificial lighting conditions at night intersections. 3) Can produce artifacts on bright objects like headlights and sky regions. 4) Static method - applies the same dehazing strength regardless of haze density. 5) No integration with object detection or tracking pipelines.

Our Proposed Improvements

1) Use Dark Channel Prior selectively only on foggy frames, detected by a weather classifier, to avoid unnecessary processing overhead. 2) Replace soft matting with guided image filtering for faster real-time processing. 3) Combine with CLAHE as a fallback for nighttime frames where dark channel prior fails. 4) Feed dehazed frames directly into YOLOv8 to quantify detection accuracy improvement. 5) Build an adaptive pipeline: weather classifier selects Dark Channel Prior (fog) vs CLAHE (night) vs no preprocessing (clear).

11 Zero-Reference Deep Curve Estimation for Low-Light Image Enhancement (Zero-DCE)

Authors: Chunle Guo, Chongyi Li, et al.
Published: 2020 | IEEE/CVF CVPR 2020
https://arxiv.org/abs/2001.06826

Abstract

Zero-DCE formulates light enhancement as image-specific curve estimation. DCE-Net (79K parameters) estimates pixel-wise curves for dynamic range adjustment. Trained with non-reference loss functions, requiring no paired training data. Achieves state-of-the-art performance while being fast and lightweight.

Introduction

Low-light enhancement is critical for nighttime applications but existing methods require expensive paired training data. Zero-DCE uses non-reference losses: spatial consistency, exposure control, color constancy, and illumination smoothness.

Technologies Used

DCE-Net (7 convolutional layers, 79K parameters), pixel-wise curve estimation, non-reference loss functions, iterative curve application, PyTorch, real-time inference.

Conclusion

Zero-DCE provides effective, data-free low-light enhancement. The lightweight architecture (79K parameters) enables real-time deployment, making it suitable for preprocessing nighttime traffic camera feeds.

Research Gap

1) Designed for general low-light enhancement, not specifically optimized for traffic/surveillance scenarios with headlight glare and streetlight artifacts. 2) Can over-enhance bright regions (headlights, traffic signals) while enhancing dark regions. 3) No evaluation of downstream task performance (e.g., how much does enhancement improve object detection accuracy?). 4) Does not handle mixed-condition frames (partially lit intersections). 5) No automatic activation - requires manual decision of when to apply.

Our Proposed Improvements

1) Integrate Zero-DCE as one module in our adaptive preprocessing pipeline, activated automatically when the weather classifier detects low-light conditions. 2) Compare Zero-DCE vs CLAHE vs no preprocessing on nighttime frames and measure the YOLOv8 mAP improvement for each. 3) Add headlight/traffic signal masking before applying enhancement to avoid over-enhancement artifacts. 4) Combine with the detection pipeline to quantify end-to-end accuracy gain from preprocessing.

12

Contrast Limited Adaptive Histogram Equalization (CLAHE)

Authors: Karel Zuiderveld

Published: 1994 | Graphics Gems IV, Academic Press, pp. 474-485

https://en.wikipedia.org/wiki/Adaptive_histogram_equalization

Abstract

CLAHE divides the image into small tiles and applies histogram equalization to each independently. A contrast limiting step clips the histogram at a predefined threshold to prevent noise amplification. Clipped pixels are redistributed uniformly across the histogram.

Introduction

Standard histogram equalization improves global contrast but amplifies noise. AHE computes local histograms but can over-enhance noise. CLAHE constrains enhancement with a clip limit, preventing noise amplification while improving local contrast.

Technologies Used

Tile-based local histogram computation, contrast clipping with redistribution, bilinear interpolation between tiles, configurable clip limit and tile grid size. Implemented in OpenCV as cv2.createCLAHE().

Conclusion

CLAHE has become the de facto standard for local contrast enhancement in surveillance and low-light applications. Its simplicity and computational efficiency make it an essential preprocessing tool for enhancing poor-visibility camera images.

Research Gap

1) A traditional image processing technique from 1994 with no learning capability - uses fixed parameters regardless of image content. 2) The clip limit and tile size must be manually tuned for each camera installation. 3) Does not handle color distortion caused by artificial lighting (sodium streetlights). 4) Can introduce block artifacts at tile boundaries in extreme cases. 5) No evaluation of impact on modern deep learning detectors like YOLO.

Our Proposed Improvements

1) Use CLAHE as the fast, lightweight preprocessing option for nighttime frames in our adaptive pipeline. 2) Auto-tune the clip limit based on frame statistics (mean brightness, histogram spread). 3) Evaluate CLAHE vs Zero-DCE vs no preprocessing and measure YOLOv8 detection accuracy for each. 4) Apply CLAHE in LAB color space (L-channel only) to avoid color distortion. 5) Combine CLAHE with detection confidence to verify that preprocessing actually improves detection.

13 Long Short-Term Memory (LSTM)

Authors: Sepp Hochreiter, Jurgen Schmidhuber

Published: November 1997 | Neural Computation, Vol. 9, No. 8, pp. 1735-1780

https://www.bioinf.jku.at/publications/older/2604.pdf

Abstract

LSTM introduces memory cells with constant error carousels that bridge time lags exceeding 1000 steps. Multiplicative gate units control information flow. Computational complexity is O(1) per time step. LSTM significantly outperforms existing RNN methods on long-time-lag tasks.

Introduction

Traditional RNNs struggle with long-term dependencies due to vanishing/exploding gradients. LSTM addresses this with memory cells containing input, output, and forget gates that control information flow, enabling learning over hundreds of time steps.

Technologies Used

Memory cells with constant error carousels, input gate, output gate, forget gate, multiplicative gating mechanisms, truncated backpropagation through time, sigmoid and tanh activations.

Conclusion

LSTM solves the vanishing gradient problem, enabling effective long-term dependency learning. It became the foundational architecture for sequence modeling, influencing speech recognition, machine translation, and time-series forecasting.

Research Gap

1) Sequential processing prevents parallelization, making training slow on large datasets. 2) Struggles with very long sequences (> 500 steps) despite the gating mechanism. 3) Standard LSTM does not incorporate external contextual features (e.g., weather, time of day) well. 4) Single-directional processing misses future context. 5) No built-in uncertainty quantification for predictions.

Our Proposed Improvements

1) Use LSTM as one of the forecasting baselines in our pipeline, comparing against TCN and XGBoost. 2) Feed additional contextual features alongside traffic counts: time of day, day of week, weather condition tag, and detection confidence. 3) Add confidence-weighted soft counts as input instead of hard integer counts to give the LSTM uncertainty information. 4) Implement Bi-LSTM for bidirectional temporal context. 5) Add prediction confidence intervals to quantify forecast uncertainty.

14 CNN-LSTM Hybrid for Short-Term Traffic Flow Prediction

Authors: L. Chen, X. Chen

Published: 2024 | IEEE Transactions on Intelligent Transportation Systems

https://ieeexplore.ieee.org/document/10443519

Abstract

Proposes a hybrid CNN-LSTM architecture where 1D-CNN layers extract local temporal patterns, which are then fed into LSTM layers for long-range sequence modeling. The hybrid captures both fine-grained local patterns and broader temporal dependencies.

Introduction

Standalone LSTM struggles with local pattern extraction while CNN lacks long-range temporal modeling. The hybrid leverages CNN for spatial/local feature extraction and LSTM for temporal sequence modeling.

Technologies Used

1D Convolutional Neural Networks, LSTM layers, fully connected output layers, sliding window approach, batch normalization, dropout regularization, Adam optimizer, PyTorch/TensorFlow.

Conclusion

CNN-LSTM achieves faster convergence and better captures periodic patterns (daily rush hours) compared to standalone models. The CNN extracts spatial correlations while LSTM handles temporal dependencies.

Research Gap

1) Uses only historical traffic flow as input - no weather or environmental context features. 2) Trained on sensor-based loop detector data, not camera-derived vehicle counts which have inherent detection noise. 3) No error propagation analysis - does not quantify how input data quality affects forecast accuracy. 4) Fixed prediction horizon - does not support multi-horizon forecasting (5/15/30 min). 5) No comparison with non-deep-learning baselines like XGBoost.

Our Proposed Improvements

1) Add weather condition, time of day, and detection confidence as additional input features to the CNN-LSTM. 2) Train on camera-derived vehicle counts from our YOLOv8+ByteTrack pipeline, which contain detection noise, and measure the impact on forecast accuracy. 3) Implement error propagation analysis to quantify how detection errors cascade into forecasting errors. 4) Support multiple prediction horizons (5, 15, 30 minutes). 5) Compare CNN-LSTM against XGBoost and standalone LSTM to identify the best model for our specific data.

15 An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling (TCN)

Authors: Shaojie Bai, J. Zico Kolter, Vladlen Koltun
Published: March 2018 | arXiv:1803.01271
https://arxiv.org/abs/1803.01271

Abstract

Proposes the Temporal Convolutional Network (TCN) using causal convolutions, dilated convolutions, and residual connections. TCN consistently outperforms LSTMs and GRUs across a broad range of tasks while being fully parallelizable and more memory-efficient.

Introduction

The deep learning community views sequence modeling as synonymous with recurrent networks. However, convolutional architectures can be equally effective. This paper provides comprehensive empirical evaluation to determine whether the bias toward recurrence is justified.

Technologies Used

Causal convolutions (no future leakage), dilated convolutions (exponentially growing receptive field), residual connections, 1D convolutions, weight normalization, dropout, flexible receptive field.

Conclusion

TCN outperforms LSTMs and GRUs while being fully parallelizable and training 2-5x faster. The results challenge the assumption that recurrence is the default for sequence modeling. TCN is increasingly preferred for real-time traffic forecasting.

Research Gap

1) Evaluated on generic sequence benchmarks, not traffic-specific time series. 2) Fixed receptive field determined at architecture design time - does not adapt to varying pattern lengths. 3) No handling of irregular time series (missing data points common in camera-based counting). 4) No multi-variate input evaluation (e.g., weather + traffic counts simultaneously). 5) Does not provide prediction uncertainty bounds.

Our Proposed Improvements

1) Evaluate TCN as an alternative to LSTM for our traffic forecasting module and compare speed vs accuracy. 2) Leverage TCN's parallelizability for faster training on our 5-minute interval time series. 3) Feed multi-variate input: soft vehicle counts + weather tag + time features + confidence scores. 4) Handle missing data points (from detection failures) with interpolation before feeding to TCN. 5) Add Monte Carlo Dropout for uncertainty estimation.

16

XGBoost: A Scalable Tree Boosting System

Authors: Tianqi Chen, Carlos Guestrin

Published: August 2016 | ACM SIGKDD (KDD) 2016

https://arxiv.org/abs/1603.02754

Abstract

XGBoost is a scalable tree boosting system with a novel sparsity-aware algorithm and weighted quantile sketch. It provides cache-aware access, data compression, and sharding to scale to billions of examples.

Introduction

Gradient boosted trees are effective but existing implementations are limited in scalability. XGBoost introduces algorithmic innovations (regularized objective, weighted quantile sketch) and systems optimizations (column-block parallel learning, cache-aware access).

Technologies Used

Gradient Boosted Decision Trees (GBDT), regularized L1+L2 objective, weighted quantile sketch, sparsity-aware split finding, column-block parallel processing, cache-aware access, out-of-core computation.

Conclusion

XGBoost achieves state-of-the-art results while being fast and scalable. It dominates Kaggle competitions. For traffic prediction, it offers high interpretability and competitive accuracy with engineered features.

Research Gap

1) Requires extensive manual feature engineering - cannot automatically extract temporal patterns from raw time series. 2) Does not model sequential dependencies like LSTM/TCN - treats each prediction independently. 3) Cannot process raw video or image data - only works on tabular features. 4) Feature importance can be misleading with correlated features. 5) No native support for time-series cross-validation.

Our Proposed Improvements

1) Use XGBoost with carefully engineered features: rolling mean counts, hour of day, day of week, weather tag, detection confidence, and lagged values. 2) Compare XGBoost against LSTM and TCN to determine if the simpler model is competitive for our specific dataset. 3) Use XGBoost's feature importance to identify which input features most affect congestion prediction. 4) Implement time-series cross-validation (expanding window) for proper evaluation. 5) The Zindi competition winners used XGBoost with engineered features, validating this approach for our exact problem.